Smart Glove: Giving Voice to the Unspoken

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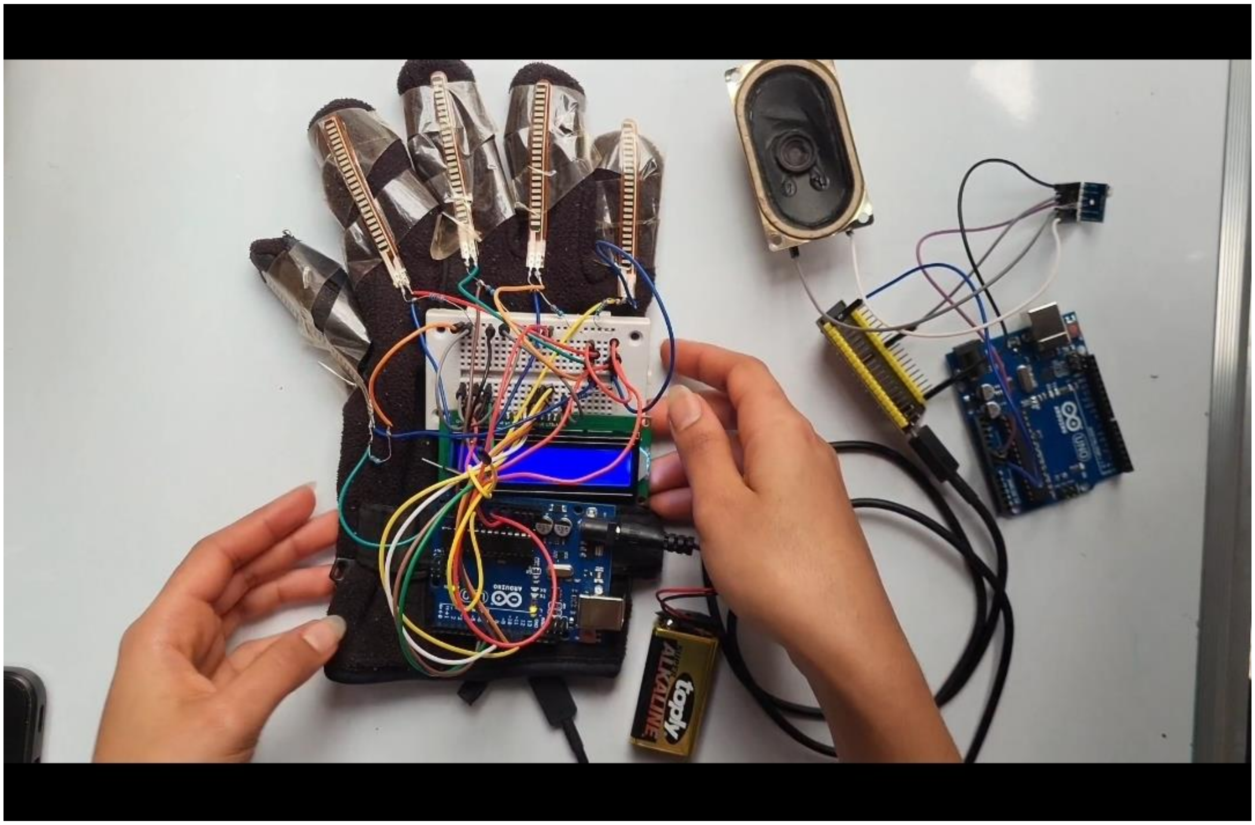
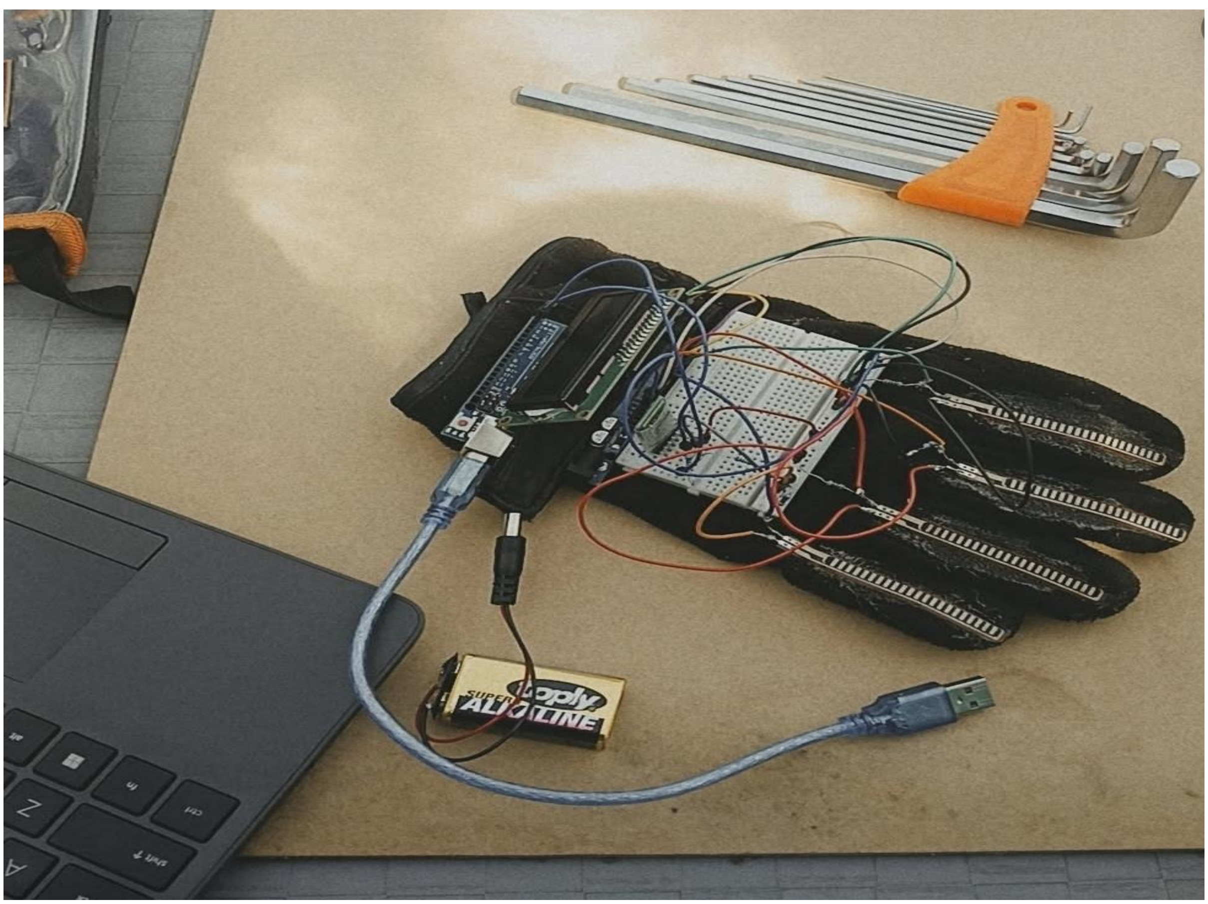
Smart Glove 1st edition



Smart Glove 2nd edition



1. Introduction

Communication is one of the most important parts of everyday life, but many people with speech impairments or disabilities face difficulties expressing their thoughts and needs. Sign language is an effective method of communication; however, not everyone understands it, which creates challenges in daily interactions. To help solve this problem, the Smart Glove: Giving Voice to the Unspoken project was developed.

The smart glove is a wearable electronic device designed to convert hand gestures into readable text and speech output. The system uses sensors and an Arduino microcontroller to detect finger movements and process specific gestures. The generated message can then be displayed on an LCD screen or transmitted to a mobile application for voice output.

This project combines electronics, programming, and assistive technology to create a simple and affordable communication solution. It also demonstrates how modern technology can be used to improve the lives of people with communication difficulties while encouraging innovation in healthcare and accessibility systems.

2. Objective

- Convert hand Gestures to readable text
- Produces voice output through a mobile app , first edition uses speaker
- Improve communication for healthcare and sign language users

3. Components Used

Components	Purpose
Arduino Uno	Microcontroller
Flex Sensors	To detect finger bending
LCD Display	Displays the words
Bluetooth Module	Send the words to the mobile app
Glove	A wearable device
Battery	Power supply

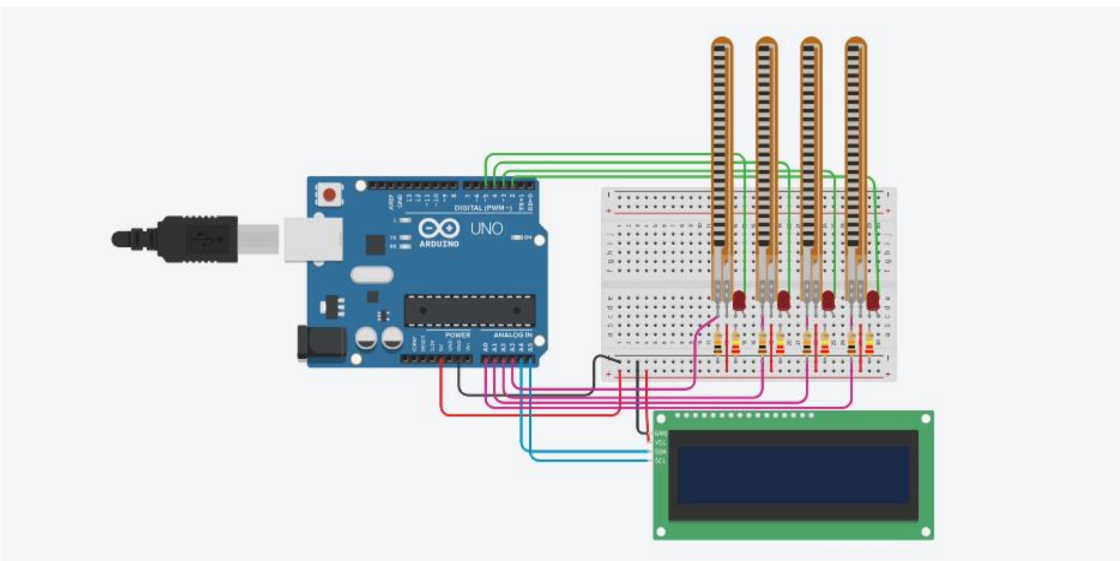
Resistors, bread board, jumper wires were also used to complete the connection

4. Working Principle

2nd edition

1. User bends fingers
2. Flex sensors detect movement
3. Arduino uno processes data
4. Message appears on LCD
5. Message sent to mobile app
6. App converts text to speech

5. Circuit Diagram (on Tinkercad)



6. Software (Arduion IDE)

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// LCD
LiquidCrystal_I2C lcd(0x27, 16, 2);

// FLEX SENSORS
int flexPins[4] = {A0, A1, A2, A3};
int threshold = 5;
```

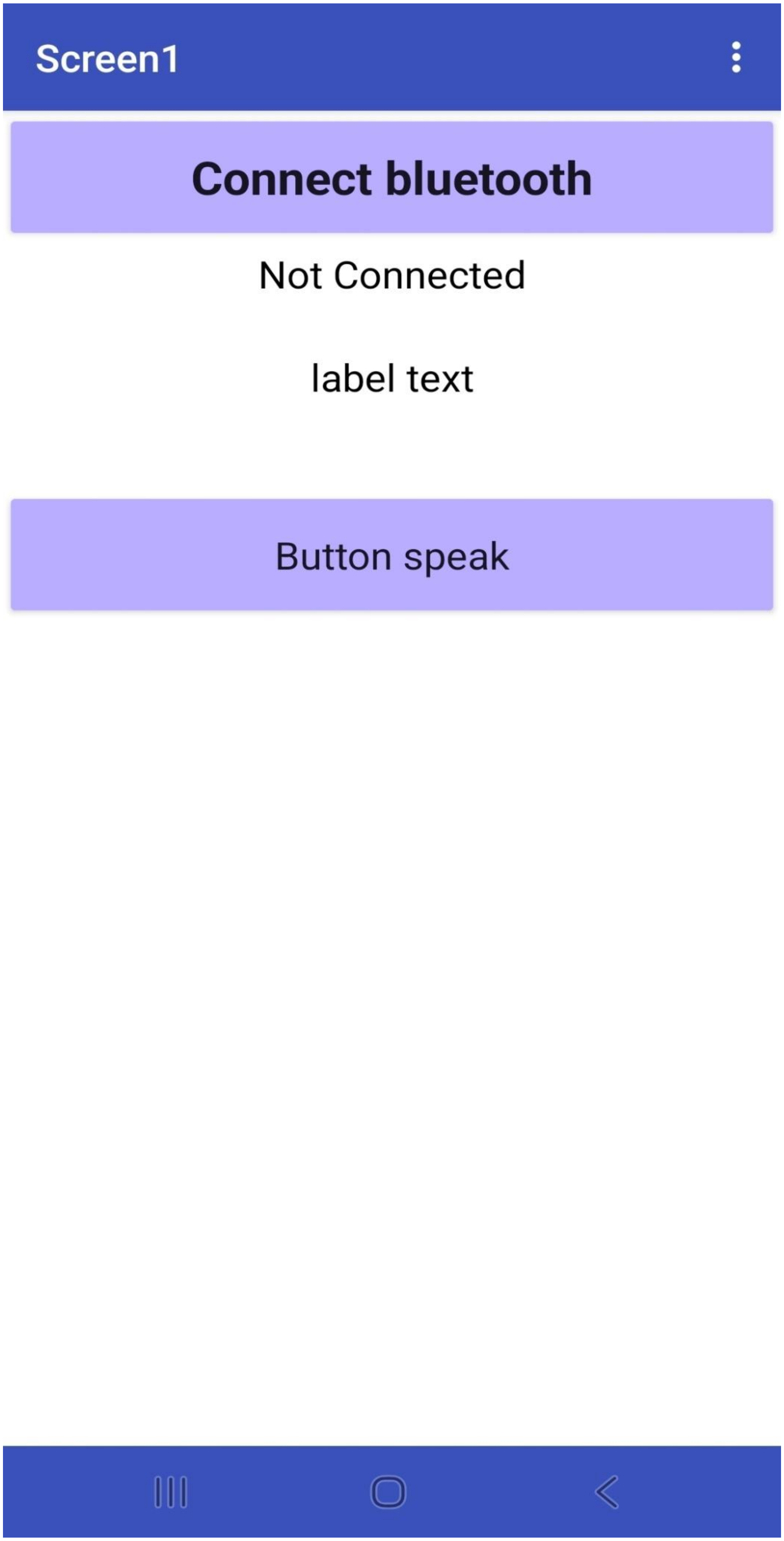
```
// MODE DETECTOR
bool isWordMode = true;

// DATA
String lastMessage = "";
String sentence = "";
int previousCode = -1;

// WORD MODE
String words[16] = {
  "", "Help", "Chest pain", "Can't breath",
  "back pain", "Fever", "Doctor", "Rash"
  "Diarrhea", "Cough", "Headache", "vomiting",
  "insomia", "bleeding", "Urgency", "Weakness"
};

// LETTER MODE
String letters[16] = {
  "", "A", "B", "C",
  "D", "E", "F", "G",
  "H", "I", "J", "K",
  "L", "M", "N", "O"
};

// GET GESTURE
int getGestureID() {
  int mask = 0;

  for (int i = 0; i < 4; i++) {
    int value = analogRead(flexPins[i]);

    if (value < threshold) {
      mask |= (1 << i);
    }
  }

  return mask;
}
```

```
void setup() {
  Serial.begin(9600);

  lcd.init();
  lcd.backlight();

  lcd.print("Smart Glove");
  delay(1500);
  lcd.clear();
}

void loop() {
  int code = getGestureID();

  // Count bent fingers
  int bentCount = 0;
  for (int i = 0; i < 4; i++) {
    if (analogRead(flexPins[i]) < threshold) {
      bentCount++;
    }
  }

  //  MODE TOGGLE
  if (bentCount == 4) {
    isWordMode = !isWordMode;

    lcd.clear();
    lcd.setCursor(0, 0);

    if (isWordMode) {
      lcd.print("Word Mode");
      Serial.println("Word Mode");
    } else {
      lcd.print("Letter Mode");
      Serial.println("Letter Mode");
      sentence = "";
    }

    delay(1000);
```

```
  previousCode = -1;
  lastMessage = "";
  return;
}

// WORD MODE
if (isWordMode) {
  String message = words[code];

  if (message != lastMessage) {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print(message);

    if (message != "") {
      Serial.println(message);
    }

    lastMessage = message;
  }
}

// LETTER MODE
else {
  if (code != 0 && code != previousCode) {

    if (code == 1) {
      sentence += " ";
    }
    else if (code == 2) {
      sentence = "";
      lcd.clear();
      Serial.println("Cleared");
      previousCode = code;
      delay(300);
      return;
    }
    else {
      sentence += letters[code];
```

```
    }

    // Display on LCD
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print(sentence.substring(0, 16));

    if (sentence.length() > 16) {
      lcd.setCursor(0, 1);
      lcd.print(sentence.substring(16, 32));
    }

    // PRINT FULL SENTENCE
    Serial.println(sentence);
  }
}

previousCode = code;
delay(150);}
```

7. Mobile Application


Screen1
Connect bluetooth
Not Connected
label text
Button speak

8. Challenge Faced

- Flex sensors threshold had trouble being determined we first made it by push buttons and try to figure it out
- The Bluetooth was not working properly and data was not being send
- The wires and their loose connectivity made it hard to actually wear the glove
- The sensors read when their not being bent and some words were randomly displayed

9. Future Improvement

The smart Glove displays certain words and is built to be applicable for health care purposes, with PCB designs implementation the circuit can be smaller and with AI recognition built into it , it can be made to actually convert sign gestures to text and by using speaker the voice output can be through it rather than the mobile application

10. Conclusion

The Smart Glove project was developed to help people with speech impairments communicate more easily through hand gestures. By using sensors, Arduino programming, and display systems, the glove is able to convert gestures into text and speech output. This project helped improve my understanding of electronics, embedded systems, and problem solving while showing how technology can be used to solve real-life challenges. In the future, the project can be improved with better gesture recognition, wireless communication, and mobile application integration to make it more accurate and user-friendly.